Chapter 19 - MOD Playback

The MOD player handles ProTracker `.mod` modules - the four-channel sample-based format that dominated Amiga music from the late 1980s. Unlike AHX, which synthesises every sample on the fly, MOD plays back pre-recorded samples stored inside the module file. The player drives the global mixer; no separate sound chip is involved.

19.1 What MOD can show

Item	Value
Channels	`4`
Sample format	Signed 8-bit, embedded in the module file
Max samples	`31` (standard) per module
Filter modes	None, Amiga 500 (4.5 kHz), Amiga 1200 (28 kHz)
File format	Standard ProTracker `.mod`

19.2 The register block

The player is a small block at `0xF0BC0`–`0xF0BD7`.

Address	Name	Purpose
`0xF0BC0`	`MOD_PLAY_PTR`	Address of the MOD data in memory.
`0xF0BC4`	`MOD_PLAY_LEN`	Length of the MOD data, in bytes.
`0xF0BC8`	`MOD_PLAY_CTRL`	Bit `0` = start, bit `1` = stop, bit `2` = loop.
`0xF0BCC`	`MOD_PLAY_STATUS`	Bit `0` = playing, bit `1` = error.
`0xF0BD0`	`MOD_FILTER_MODEL`	Output filter (`0` = none, `1` = A500, `2` = A1200).
`0xF0BD4`	`MOD_POSITION`	Current song position (read-only).

19.2.1 The filter models

The Amiga 500's analogue output filter had a corner at about `4.5` kHz, which gave its music a warmer, more rounded sound; the Amiga 1200's corner was at about `28` kHz, much closer to flat. `MOD_FILTER_MODEL` lets you choose which one (or neither) to apply, so the same `.mod` can be heard as either machine would have played it.

19.3 Programming model

To play a module:

1. Load the file into memory.
2. Write the base address to `MOD_PLAY_PTR` and the size to `MOD_PLAY_LEN`.
3. (Optional) write a filter model to `MOD_FILTER_MODEL`.
4. Write `1` to `MOD_PLAY_CTRL`.

To stop: write `2`. To loop: combine `1 | 4` (start + loop).

19.4 BASIC keywords

There is no dedicated `MOD` keyword in BASIC. To play a `.mod` file from disk in one step, use the `SOUND PLAY` keyword (Chapter 22); the media loader picks the MOD engine from the `.mod` extension. To drive the player from BASIC directly, use `POKE` against the registers above.

```
10 POKE &H000F0800, 1                : REM AUDIO_CTRL = on
20 BLOAD "song.mod", &H100000
30 POKE &H000F0BC0, &H00100000        : REM MOD_PLAY_PTR
40 POKE &H000F0BC4, 65536             : REM MOD_PLAY_LEN (size in bytes)
50 POKE &H000F0BD0, 1                 : REM filter model = A500
60 POKE &H000F0BC8, 1                 : REM start
```

19.5 Putting it together

MOD playback is the simplest way to add high-quality background music to a program without writing any synthesis code. The module file carries the samples, the patterns, and the order; the player handles everything else.

The next chapter covers the WAV / sample player, used for short sound effects and digitised speech.